EE605 Quiz 4 Solutions

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1 Question 1

It is trivial to show that the set of such codewords (call it C) is a linear code, and that its length is $q+1$. See that a generator matrix for C is

$$\begin{pmatrix} 1 & 1 & \dots & 1 & 0 \\ \alpha_1 & \alpha_2 & \dots & \alpha_q & 0 \\ \alpha_1^2 & \alpha_2^2 & \dots & \alpha_q^2 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ \alpha_1^{k-2} & \alpha_2^{k-2} & \dots & \alpha_q^{k-2} & 0 \\ \alpha_1^{k-1} & \alpha_2^{k-1} & \dots & \alpha_q^{k-1} & 1 \end{pmatrix}$$

where α_i are the distinct elements of $\mathbb{F}_q$. It was already shown in quiz 3 that the code generated by such a matrix is MDS. This code is permutation-equivalent to that code, and any code permutation-equivalent to an MDS code is also MDS. For the sake of completeness, that proof is reproduced verbatim below. Let $C' = C_{\{q+1\}}$ i.e. the shortening of C at the last coordinate. Then clearly, a parity check matrix for C' is

$$\begin{pmatrix} 1 & 1 & \dots & 1 & 1 \\ \alpha_1 & \alpha_2 & \dots & \alpha_{q-1} & \alpha_q \\ \alpha_1^2 & \alpha_2^2 & \dots & \alpha_{q-1}^2 & \alpha_q^2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ \alpha_1^{q-k} & \alpha_2^{q-k} & \dots & \alpha_{q-1}^{q-k} & \alpha_q^{q-k} \end{pmatrix}$$

where $\alpha_1, \dots, \alpha_{q-1}$ are the nonzero elements of $\mathbb{F}_q$ and α_q is the zero element of $\mathbb{F}_q$. Note then that a generator matrix for $C'^{\perp}$ is H', and H' has order $q \times (q-k+1)$ so $C'^{\perp}$ is a $(q, q-k+1)-$Reed-Solomon code $\Rightarrow C'^{\perp}$ is MDS $\Rightarrow C'$ is MDS $\Rightarrow d(C') = q-k+2 \Rightarrow$ there is a set of $q-k+2$ columns of H' which is linearly dependent, and every set of $q-k+1$ columns of H' is linearly independent. So there is a set of $q-k+2$ columns of H which is linearly dependent (since every column of H' is also a column of H). Now take a set of $q-k+1$ columns from H. If all are also columns of H', we are done. If not, the set must be some $q-k$ columns from H' and the column $(0\ 0\ \dots\ 0\ 1)^T$. Make these columns into a square matrix. It will look like below.

$$\begin{pmatrix} 1 & 1 & \dots & 1 & 0 \\ \alpha_{i_1} & \alpha_{i_2} & \dots & \alpha_{i_{q-k}} & 0 \\ \alpha_{i_1}^2 & \alpha_{i_2}^2 & \dots & \alpha_{i_{q-k}}^2 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ \alpha_{i_1}^{q-k-1} & \alpha_{i_2}^{q-k-1} & \dots & \alpha_{i_{q-k}}^{q-k-1} & 0 \\ \alpha_{i_1}^{q-k} & \alpha_{i_2}^{q-k} & \dots & \alpha_{i_{q-k}}^{q-k} & 1 \end{pmatrix}$$

where $i_1, i_2, \dots, i_{q-k}$ denote the indices of the chosen $q-k$ columns of H'. Note that this is a square matrix, so its columns are linearly independent iff its determinant is 0. Expanding the determinant along the last column, we get the value of its determinant to be

$$\det\left(\begin{pmatrix} 1 & 1 & \dots & 1 \\ \alpha_{i_1} & \alpha_{i_2} & \dots & \alpha_{i_{q-k}} \\ \alpha_{i_1}^2 & \alpha_{i_2}^2 & \dots & \alpha_{i_{q-k}}^2 \\ \vdots & \vdots & \ddots & \vdots \\ \alpha_{i_1}^{q-k-1} & \alpha_{i_2}^{q-k-1} & \dots & \alpha_{i_{q-k}}^{q-k-1} \end{pmatrix}\right)$$

which is a Vandermonde matrix. Since all of $\alpha_{i_1}, \alpha_{i_2}, \ldots, \alpha_{i_{q-k}}$ are distinct, this determinant is nonzero. We are done.

2 Question 2

The outer code is a Reed Solomon code $C_{out} \subseteq F_{2^t}^{2^t}$, $t > 0$, with rate ϵ. From this, we get $n_{out} = 2^t, k_{out} = \epsilon n_{out} = \epsilon 2^t$, and $d_{out} = n_{out} - k_{out} + 1 = (1-\epsilon)2^t + 1$.

The inner code is a binary Hadamard code with parameters $n_{in} = 2^t$, and $k_{in} = \log_2 |F_{2^t}| = t$. All non-zero codewords have Hamming weight 2^{t-1}, thus $d_{in} = 2^{t-1}$.

The parameters of the concatenated code C are $n = n_{out} \cdot n_{in} = 2^{2t}, k = k_{out} \cdot k_{in} = \epsilon 2^t t$, and $d = d_{out} \cdot d_{in} = ((1-\epsilon)2^t + 1)2^{t-1}$.

3 Question 3

First and foremost, we need to check for well-definition at $x = 0$ and $x = 1$. See that

$$\lim_{x \to 0^+} x \log_q(x) = \frac{1}{\ln(q)} \lim_{x \to 0^+} \frac{\ln(x)}{\frac{1}{x}} = \frac{1}{\ln(q)} \lim_{x \to 0^+} \frac{\frac{1}{x}}{\frac{-1}{x^2}} = \frac{-1}{\ln(q)} \lim_{x \to 0^+} x = 0$$

by L'Hôpital's rule. So similarly, $\lim\limits_{x \to 1^-} (1-x) \log_q(1-x) = 0$. Define thus for $q \in \mathbb{N} \setminus \{1\}$ and $x \in [0,1]$

$$H_q(x) = \begin{cases} x \log_q(q-1) - x \log_q(x) - (1-x) \log_q(1-x) & \text{, if } x \in (0,1) \\ 0, & \text{, if } x = 0 \\ \log_q(q-1) & \text{, if } x = 1 \end{cases}$$

Having done this, H_q is continuous on $[0,1]$ and differentiable on $(0,1)$.

3.a

For $x \in (0,1)$,

$$\begin{aligned} \frac{dH_q(x)}{dx} &= \log_q(q-1) - \log_q(x) - x\frac{1}{x\ln(q)} + \log_q(1-x) + (1-x)\frac{1}{(1-x)\ln(q)} \\ &= \log_q(q-1) - \log_q(x) + \log_q(1-x) \\ &= \log_q\left(\frac{(q-1)(1-x)}{x}\right) \end{aligned}$$

So

$$\begin{aligned} \frac{dH_q(x)}{dx} = 0 \text{ at } \frac{(q-1)(1-x)}{x} &= 1 \\ \text{i.e. at } q - 1 &= qx \\ \text{i.e. at } x &= \frac{q-1}{q} \end{aligned}$$

For $x \in \left(0, \frac{q-1}{q}\right)$, we have $\frac{(q-1)(1-x)}{x} > 1 \Rightarrow \frac{dH_q(x)}{dx} > 0 \Rightarrow H_q$ is strictly increasing in $\left(0, \frac{q-1}{q}\right)$. For $x \in \left(\frac{q-1}{q}, 1\right)$, we have $\frac{(q-1)(1-x)}{x} < 1 \Rightarrow \frac{dH_q(x)}{dx} < 0 \Rightarrow H_q$ is strictly decreasing in $\left(\frac{q-1}{q}, 1\right)$. So H_q must have a local maximum at $x = \frac{q-1}{q}$. But H_q is continuous on $[0,1]$, so this must be the (unique) global maximum. This answers b first. The maximum value of H_q in $[0,1]$ is thus at $x = \frac{q-1}{q}$, and that is

$$\begin{aligned} H_q\left(\frac{q-1}{q}\right) &= \frac{q-1}{q}\log_q(q-1) - \frac{q-1}{q}\log_q\left(\frac{q-1}{q}\right) - \frac{1}{q}\log_q\left(\frac{1}{q}\right) \\ &= \frac{q-1}{q}\log_q(q-1) - \frac{q-1}{q}\log_q(q-1) + \frac{q-1}{q}\log_q(q) - \frac{1}{q}\log_q\left(\frac{1}{q}\right) \\ &= \frac{q-1}{q} + \frac{1}{q} = 1 \end{aligned}$$

3.b

Answered already in (a).

3.c

Define $g(x) = H_q(x) - x$. Then clearly

$$g'(x) = \log_q\left(\frac{(q-1)(1-x)}{x}\right) - 1 = \log_q\left(\frac{(q-1)(1-x)}{qx}\right)$$

which is zero at $(q-1)(1-x) = qx \Rightarrow x = \frac{q-1}{2q-1}$. Note that this value of x is in strictly in between 0 and $1-\frac{1}{q}$ for $q \in \mathbb{N} \setminus \{1\}$. Further note that for $x \in \left(0, \frac{q-1}{2q-1}\right)$, $\frac{(q-1)(1-x)}{qx} > 1 \Rightarrow g'(x) > 0 \Rightarrow g$ is strictly increasing in $\left(0, \frac{q-1}{2q-1}\right)$. Similarly, we get that g is strictly decreasing in $\left(\frac{q-1}{2q-1}, 1-\frac{1}{q}\right]$. So it must be that the minimum of g in $\left[0, 1-\frac{1}{q}\right]$ occurs at one of the boundaries i.e. either at 0 or at $1-\frac{1}{q}$. $g(0) = H_q(0) - 0 = 0$ and $g\left(1-\frac{1}{q}\right) = H_q\left(1-\frac{1}{q}\right) - \left(1-\frac{1}{q}\right) = 1 - 1 + \frac{1}{q} = \frac{1}{q} > 0$. Hence $g(x) \geq 0 \, \forall x \in \left[0, 1-\frac{1}{q}\right]$. We are done.

4 Question 4

4.a

A matrix is d-disjunct if no column is covered by the OR of any other d columns. Given binary test matrix $M = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \end{bmatrix}$. Note that for any column (say a), there always exists some row i s.t. $a_i = 1$, and $b_i = 0$, where b is another column out of the remaining 3 columns. Therefore $a_i \leq b_i \; \forall \; i \in [5]$ does not hold for any a and b pair, hence M is 1-disjunct.

Next, consider OR of columns 2 and 4, resulting in the all-ones vector, which covers any other column. Hence M is not 2-disjunct, and largest d for which M is d-disjunct is 1.

4.b

A matrix M is 2-separable if all subsets of columns of size ≤ 2 give distinct OR outcomes. Consider individual columns and OR output of all distinct pairs of columns.

The resulting outcomes are $\begin{bmatrix}1\\0\\0\\1\\0\end{bmatrix}, \begin{bmatrix}0\\1\\0\\1\\1\end{bmatrix}, \begin{bmatrix}0\\0\\1\\0\\1\end{bmatrix}, \begin{bmatrix}1\\1\\1\\0\\0\end{bmatrix}, \begin{bmatrix}1\\1\\0\\1\\1\end{bmatrix}, \begin{bmatrix}1\\0\\1\\1\\1\end{bmatrix}, \begin{bmatrix}1\\1\\1\\1\\0\end{bmatrix}, \begin{bmatrix}0\\1\\1\\1\\1\end{bmatrix}, \begin{bmatrix}1\\1\\1\\1\\1\end{bmatrix}, \begin{bmatrix}1\\1\\1\\0\\1\end{bmatrix}$ which are all distinct. Hence, M is 2-separable.

4.c

Let the four columns be denoted by c_1, c_2, c_3, c_4. $|D| \leq 1$ is to be recovered from $y = \begin{bmatrix}1\\0\\0\\1\\0\end{bmatrix}$ using simple elimination decoding rule. An item j is defective if all tests that include the item are positive, i.e. $c_j \leq y$.

- Consider item 1. $c_1 \leq y$, so item 1 is kept.
- For each of the items 2,3, and 4, there exists at least one test containing the item that is negative, i.e. $c_j \nleq y \; \forall \; j \in \{2, 3, 4\}$. Therefore these items are eliminated, resulting in $D = \{1\}$.

It can also be noted that since $y = c_1$, any column c_j satisfying $c_j \leq y$ must also satisfy $c_j \leq c_1$. However, since the matrix M is 1-disjunct, no column can be covered by any other column. Therefore, for all $j \in \{2, 3, 4\}$, $c_j \not\leq c_1$, and hence $c_j \not\leq y$. On the other hand, $c_1 \leq y$ holds trivially since $c_1 = y$. Therefore, item 1 is the only possible defective item, and $D = \{1\}$ is concluded.

4.d

$|D| \leq 2;\ y = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 0 \end{bmatrix}$. No individual column matches y. Considering pairs, $c_1 \vee c_2 = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 1 \\ 1 \end{bmatrix}, c_1 \vee c_3 = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 1 \\ 1 \end{bmatrix}, c_1 \vee c_4 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 0 \end{bmatrix}, c_2 \vee c_3 = \begin{bmatrix} 0 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, c_2 \vee c_4 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, c_3 \vee c_4 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \\ 1 \end{bmatrix}$. Note that only $c_1 \vee c_4$ matches y. Therefore, $D = \{1, 4\}$.